

SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone Project

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GitHub: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor>

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Executive Summary

Objective

- Predict whether SpaceX Falcon 9 first stage will land successfully, directly impacting launch cost estimation (\$62M vs. \$165M+)
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Data & Methodology

- Collected 90 Falcon 9 launch records (2010-2020) via SpaceX REST API and Wikipedia web scraping
- Performed data wrangling, EDA with visualization & SQL, interactive analytics with Folium & Plotly Dash
- Built and tuned 4 classification models: Logistic Regression, SVM, Decision Tree, KNN
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Key Results

- All 4 models achieved 83.33% test accuracy with identical confusion matrices
- Decision Tree achieved highest cross-validation accuracy at 88.93%
- Launch success rate improved from 0% (2010) to ~95%+ (2020), driven by technology evolution
- KSC LC-39A emerged as the most successful launch site; Falcon 9 Block 5 is the most capable booster
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Business Impact

- Accurate landing prediction enables competitive bidding against SpaceX by estimating true launch costs

Introduction

Project Background

- SpaceX revolutionized the space industry with reusable Falcon 9 rockets
- A Falcon 9 launch costs approximately \$62 million vs. \$165 million+ for competitors
- The cost advantage comes primarily from first-stage booster reuse
- If we can predict whether the first stage will land, we can estimate the launch cost
-

Research Questions

- What factors determine whether the Falcon 9 first stage will land successfully?
- How has launch success rate evolved over time and across different booster versions?
- Which launch sites and orbit types have the highest success rates?
- Can we build a machine learning model to accurately predict landing outcomes?
-

Stakeholders

- Competing space launch providers seeking to bid against SpaceX
- Space agencies and commercial satellite operators planning missions
- Aerospace analysts and investors evaluating SpaceX's competitive positioning

Section 1: Methodology

Methodology Overview

Data Collection

- SpaceX REST API (v4) — collected launch records with rocket, payload, core, and launchpad data
- Wikipedia web scraping (BeautifulSoup) — 121 Falcon 9/Heavy launch records with 11 attributes
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Data Wrangling

- Filtered Falcon 9 launches only (90 records), handled missing values, created binary Class label
- One-hot encoded categorical features, resulting in 83-column feature matrix
-

Exploratory Data Analysis

- Visualization with Matplotlib & Seaborn — scatter plots, bar charts, line charts
- SQL analysis with SQLite — queried launch sites, payload stats, landing outcomes, trends
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Interactive Visual Analytics

- Folium maps — launch site locations, success/failure markers, proximity analysis
- Plotly Dash — interactive dashboard with pie charts, scatter plots, dropdown & slider filters
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Predictive Analysis

- Built 4 classification models with GridSearchCV hyperparameter tuning (10-fold CV)
- Models: Logistic Regression, SVM, Decision Tree, KNN

Data Collection

Two complementary data collection approaches were used:

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1. SpaceX REST API (v4)

- Endpoint: `api.spacexdata.com/v4/launches/past`
- Resolved nested IDs via sub-API calls to `/rockets`, `/launchpads`, `/payloads`, `/cores`
- Extracted 17 features per launch including payload mass, orbit, grid fins, reuse count, etc.
-

2. Wikipedia Web Scraping

- Source: List of Falcon 9 and Falcon Heavy launches (Wikipedia)
- Used BeautifulSoup + requests to parse HTML tables
- Extracted 11 features: flight number, date, time, booster version, launch site, and more
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Result: 90 Falcon 9 launch records spanning June 2010 to November 2020

Data Collection — SpaceX API

API Endpoints Used:

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Endpoint	Purpose	Key Data Extracted
/v4/launches/past	All past launches	Flight number, date, rocket/payload/core IDs
/v4/rockets/{id}	Rocket details	Booster name (Falcon 9 vs Falcon 1)
/v4/launchpads/{id}	Launch pad info	Site name, longitude, latitude
/v4/payloads/{id}	Payload details	Mass (kg), target orbit
/v4/cores/{id}	Core booster info	Block version, reuse count, serial, landing outcome

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Processing Pipeline:

- Fetched raw JSON → Flattened with `pd.json_normalize` → Filtered to single-core/payload rows
- Resolved IDs via helper functions → Removed Falcon 1 launches → Reset flight numbers
- Imputed 5 missing PayloadMass values with column mean
- Exported clean dataset as `dataset_part_1.csv` (90 rows × 17 columns)
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Notebook: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/jupyter-labs-spacex-data-collection-api-v2.ipynb>

Data Collection — Web Scraping

Source & Tools:

- Website: Wikipedia — List of Falcon 9 and Falcon Heavy launches (static snapshot, June 2021)
- Libraries: requests (HTTP), BeautifulSoup (HTML parsing), unicodedata (Unicode normalization)
-

Extraction Process:

- Fetched HTML page with custom User-Agent header
- Identified launch data tables by CSS class 'wikitable plainrowheaders collapsible'
- Parsed each row using helper functions: `date_time()`, `booster_version()`, `landing_status()`, `get_mass()`
- Extracted 11 columns: Flight No., Date, Time, Booster Version, Launch Site, Payload, etc.
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Data Quality Notes:

- 121 launch records extracted across Falcon 9 and Falcon Heavy missions
- Applied Unicode NFKD normalization for payload mass strings with non-breaking spaces
- Output saved as `spacex_web_scraped.csv`
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Notebook: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/jupyter-labs-webscraping.ipynb>

Data Wrangling

Objective: Transform raw data into ML-ready features with a binary target variable

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Key Steps:

- Loaded dataset_part_1.csv (90 rows × 17 columns from API collection)
- Analyzed missing values: LandingPad had 28.89% nulls (no landing attempt on early flights)
- Identified 7 numerical and 7 categorical features plus 3 boolean columns
-

Target Variable (Class) Creation:

- Converted 8 distinct Outcome values into binary: Class = 1 (success) or Class = 0 (failure)
- Success (Class=1): True ASDS, True RTLS, True Ocean — booster landed successfully
- Failure (Class=0): False ASDS, False Ocean, False RTLS, None ASDS, None None — did not land
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Feature Engineering:

- One-hot encoded: Orbit, LaunchSite, LandingPad, Serial → 83 feature columns
- Cast all features to float64 for ML compatibility
- Overall success rate: 66.67% (60 successes out of 90 launches)
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Notebook: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/labs-jupyter-spacex-Data%20wrangling-v2.ipynb>

EDA with Data Visualization

Visualizations Created (Matplotlib & Seaborn):

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- Flight Number vs. Payload Mass (scatter) — Higher flight numbers show more successes
- Flight Number vs. Launch Site (scatter) — Later missions trend toward success across all sites
- Payload Mass vs. Launch Site (scatter) — VAFB SLC-4E has no heavy payload launches (>10,000 kg)
- Success Rate by Orbit Type (bar chart) — ES-L1, GEO, HEO, SSO achieve ~100% success
- Flight Number vs. Orbit Type (scatter) — LEO success correlates with flight number; GTO shows no pattern
- Payload Mass vs. Orbit Type (scatter) — Heavy payloads succeed in Polar, LEO, ISS orbits
- Launch Success Yearly Trend (line chart) — Consistent improvement from 2013 to 2020
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Key Insight: Success rate climbed from 0% in 2010 to ~95%+ by 2020, with a sharp increase after 2015

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Notebook: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/jupyter-labs-eda-dataviz-v2.ipynb>

EDA with SQL

Database: SQLite | 101 launch records loaded into SPACEXTABLE

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SQL Queries Performed:

- Identified 4 unique launch sites: CCAFS LC-40, CCAFS SLC-40, VAFB SLC-4E, KSC LC-39A
- Calculated total payload for NASA (CRS) missions: 45,596 kg
- Average payload mass for F9 v1.1 booster: 2,928.4 kg
- First successful ground pad landing: December 22, 2015 (historic Orbcomm-2 mission)
- Found 4 boosters with successful drone ship landing (payload 4,000-6,000 kg) — all F9 FT variants
- Mission outcomes: 98 successes, 1 failure, 2 qualified successes out of 101 total
- Maximum payload carried by 12 different Falcon 9 Block 5 boosters (all reusable cores)
- 2015 had 2 failed drone ship landings — both F9 v1.1 at CCAFS LC-40 (Jan & Apr)
- Landing outcome rankings (2010-2017): 'No attempt' most common (10), drone ship 50% success rate
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Notebook: https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Interactive Visual Analytics — Folium Maps

Launch Sites Map:

- Mapped all 4 SpaceX launch sites with labeled markers and proximity circles
- Florida sites (28.56°N): CCAFS LC-40, CCAFS SLC-40, KSC LC-39A — clustered within ~8 km
- California site (34.63°N): VAFB SLC-4E — for polar orbit launches
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Success/Failure Marker Clusters:

- Green markers = successful landings, Red markers = failed landings
- KSC LC-39A shows predominantly green markers (high success rate)
- CCAFS SLC-40 shows mixed green/red markers (variable success rate)
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Proximity Analysis (CCAFS SLC-40):

- Coastline: ~0.87 km (safety — rockets launch over open ocean)
- Highway: ~0.61 km (logistics access)
- Railway: ~1.2 km (component transportation)
- Nearest city: ~19.1 km (populated area safety buffer)
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Key Finding: All sites are coastal, at low latitudes, near transport infrastructure, and safely distant from cities

Interactive Visual Analytics — Plotly Dash Dashboard

Dashboard Components:

- Launch Site Dropdown — filter by individual site or view all sites
- Payload Range Slider — filter launches by payload mass (0-10,000 kg)
- Interactive Pie Chart — success distribution by site or success/failure split per site
- Interactive Scatter Plot — payload mass vs. launch outcome, color-coded by booster version
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Dashboard Insights:

- All Sites view: Shows which sites contribute most to total successes
- Per-site view: KSC LC-39A shows the highest success ratio
- Booster evolution visible: v1.0 and v1.1 associated with failures; FT, B4, B5 with successes
- Payload analysis: Light and medium payloads have mixed outcomes; pattern varies by site
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Dashboard Code: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/spacex-dash-app.py>

Predictive Analysis — Methodology

Data Preparation:

- Features: 83 columns (one-hot encoded) including FlightNumber, PayloadMass, Orbit, LaunchSite, etc.
- Target: Binary Class (0 = failed landing, 1 = successful landing)
- StandardScaler applied to normalize features
- Train/Test split: 80/20 (72 training, 18 test samples, random_state=2)
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Classification Models & Hyperparameter Tuning (GridSearchCV, cv=10):

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Model	Hyperparameter Grid	Best Parameters
Logistic Regression	C: [0.01, 0.1, 1], penalty: l2	C=0.01, penalty=l2, solver=lbfgs
SVM (SVC)	kernel: linear/rbf/poly/sigmoid C: logspace(-3,3,5)	C=1.0, gamma=0.0316, kernel=sigmoid
Decision Tree	criterion: gini/entropy max_depth: 2-18, splitter: best/random	criterion=gini, max_depth=2 min_samples_leaf=2
KNN	n_neighbors: 1-10 algorithm: auto/ball_tree/kd_tree/brute	n_neighbors=10, p=1 algorithm=auto

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Notebook: <https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb>

Section 2: EDA Results — Visualization

Flight Number vs. Launch Outcome

Chart: Scatter plot — Flight Number vs. Payload Mass, colored by Class (0/1)

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Key Findings:

- As flight number increases, success rate increases significantly
- Early flights (1-20) show predominantly red markers (failures)
- Later flights (60-90) show predominantly green markers (successes)
- This reflects SpaceX's learning curve and technology improvements over time

Section 2

Insights drawn from EDA

Payload Mass Relationship:

- Heavier payloads (>10,000 kg) show mixed results
- Medium payloads (2,000-6,000 kg) have the most data points and generally high success rates
- The relationship between payload and success is not strictly linear — other factors contribute

Launch Site vs. Payload Mass Analysis

Charts: Scatter plots — Flight Number vs. Launch Site & Payload Mass vs. Launch Site

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Launch Site Observations:

- CCAFS SLC-40: Highest volume site (55 launches), mixed success rates
- KSC LC-39A: Strong success record with 22 launches, handles heavy payloads well
- VAFB SLC-4E: 13 launches, specialized for polar/SSO orbits, NO heavy payloads (>10,000 kg)

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Site-Specific Insights:

- KSC LC-39A handles the widest payload mass range with consistently high success
- VAFB is limited to lighter payloads due to its polar orbit mission profile
- All sites show improvement in success rates over time (higher flight numbers)

Success Rate by Orbit Type

Chart: Bar chart — Mean Class (success rate) by Orbit

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Orbit	Success Rate	# Launches	Notes
ES-L1	100%	1	Sun-Earth Lagrange point
GEO	100%	1	Geostationary orbit
HEO	100%	1	Highly elliptical orbit
SSO	100%	5	Sun-synchronous orbit
VLEO	100%	14	Very low Earth orbit (Starlink)
ISS	95.2%	21	International Space Station resupply
PO	88.9%	9	Polar orbit
LEO	85.7%	7	Low Earth orbit
MEO	66.7%	3	Medium Earth orbit
GTO	51.9%	27	Geostationary transfer — most challenging
SO	0%	1	Sub-orbital test flight

Key Insight: GTO is the most common yet most challenging orbit with ~52% success rate

Launch Success Yearly Trend

Chart: Line chart — Average success rate by year (2010-2020)

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Year-by-Year Progression:

- 2010-2013: 0% success rate — no landing attempts during early Falcon 9 missions
- 2014: First partial successes — controlled ocean landings
- 2015: Historic milestone — first successful ground pad landing (Dec 22, RTLS at LZ-1)
- 2016: Success rate jumped to ~60% — first drone ship landing achieved
- 2017: Continued improvement, ~75% success rate
- 2018-2020: Success rate exceeded 90%, approaching near-perfect reliability

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Inflection Point: 2015 marks the transition from experimental to operational landing capability

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Driving Factors: Grid fins, landing legs, improved guidance software, Block 5 booster design

Section 2: EDA Results — SQL Analysis

SQL: Unique Launch Sites

Query: SELECT DISTINCT Launch_Site FROM SPACEXTABLE

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Launch Site	Location	Primary Use
CCAFS LC-40	Cape Canaveral, Florida	Early Falcon 9 launches, GTO/LEO missions
CCAFS SLC-40	Cape Canaveral, Florida	Falcon 9 launches after LC-40 rebuild
KSC LC-39A	Kennedy Space Center, Florida	Heavy payloads, crewed missions
VAFB SLC-4E	Vandenberg AFB, California	Polar/SSO orbit missions

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3 sites in Florida (east coast) for equatorial/ISS orbits, 1 site in California for polar orbits

- Florida sites leverage Earth's rotational velocity for eastward launches
- VAFB enables southward launches over the Pacific for polar orbits

SQL: Payload Mass Statistics

Total Payload Mass — NASA (CRS) Missions

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Query: `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'`

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Result: 45,596 kg total payload delivered to the International Space Station

- NASA CRS (Commercial Resupply Services) represents SpaceX's most consistent customer

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Average Payload Mass — F9 v1.1 Booster

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Query: `SELECT AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version = 'F9 v1.1'`

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Result: 2,928.4 kg average payload per F9 v1.1 launch

- F9 v1.1 was the transitional booster version before Full Thrust (FT) upgrade

SQL: First Successful Ground Landing Date

Query: SELECT MIN(Date) WHERE Landing_Outcome = 'Success (ground pad)'

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Result: December 22, 2015

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Historical Significance:

- First-ever orbital-class rocket booster to land vertically after delivering payload to orbit
- Mission: Orbcomm-2 (11 OG2 satellites to LEO)
- Landing site: Landing Zone 1 (LZ-1), Cape Canaveral
- This milestone proved first-stage reusability was commercially viable
- Marked the beginning of SpaceX's era of routine booster recovery

SQL: Successful Drone Ship Landings (Payload 4,000-6,000 kg)

Query: WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS BETWEEN 4000 AND 6000

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Booster Version	Type
F9 FT B1022	Falcon 9 Full Thrust
F9 FT B1026	Falcon 9 Full Thrust
F9 FT B1021.2	Falcon 9 Full Thrust (2nd flight)
F9 FT B1031.2	Falcon 9 Full Thrust (2nd flight)

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All successful mid-range payload drone ship landings were Falcon 9 Full Thrust variants

- Two of the four boosters (.2 suffix) were on their second flight, demonstrating reusability
- The FT variant introduced significant performance improvements enabling ASDS landings

SQL: Total Successful and Failure Mission Outcomes

Query: SELECT Mission_Outcome, COUNT(*) GROUP BY Mission_Outcome

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Mission Outcome	Count	Percentage
Success	98	97.0%
Success (payload status unclear)	1	1.0%
Success	1	1.0%
Failure (in flight)	1	1.0%

98% Mission Success Rate — Only 1 in-flight failure out of 101 total missions

- The single failure was SpaceX CRS-7 (June 2015) due to a strut failure in the upper stage
- Mission success is distinct from landing success — most missions deliver payload even when booster doesn't land

SQL: Boosters Carrying Maximum Payload

Query: `SELECT Booster_Version WHERE PAYLOAD_MASS = (SELECT MAX(PAYLOAD_MASS))`

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12 boosters carried the maximum payload mass — all Falcon 9 Block 5:

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Booster	Booster	Booster	Booster
F9 B5 B1048.4	F9 B5 B1049.4	F9 B5 B1051.3	F9 B5 B1056.4
F9 B5 B1048.5	F9 B5 B1051.4	F9 B5 B1049.5	F9 B5 B1060.2
F9 B5 B1058.3	F9 B5 B1051.6	F9 B5 B1060.3	F9 B5 B1049.7

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All are Block 5 — the most capable & reusable variant. Suffix numbers indicate reflights (.4 = 4th flight, .7 = 7th flight)

- B1049 flew 7 times, B1051 flew 6 times — demonstrating remarkable booster durability
- All maximum payload missions were Starlink deployments (~15,600 kg to VLEO)

SQL: 2015 Failed Drone Ship Landings

Query: WHERE Landing_Outcome = 'Failure (drone ship)' AND Year = '2015'

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Month	Booster Version	Launch Site	Landing Outcome
January 2015	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
April 2015	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

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Context:

- Both failures involved F9 v1.1 boosters — the version before Full Thrust improvements
- Both launched from CCAFS LC-40 — SpaceX's primary East Coast pad at the time
- These early drone ship landing attempts were critical learning experiences
- The January attempt (CRS-5) nearly succeeded but tipped over on the deck
- By December 2015, SpaceX achieved its first successful landing (ground pad)
- By April 2016, the first successful drone ship landing was achieved with the upgraded FT booster

SQL: Landing Outcome Rankings (2010-2017)

Query: SELECT Landing_Outcome, COUNT(*) WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY COUNT DESC

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Rank	Landing Outcome	Count	Percentage
1	No attempt	10	32.3%
2	Success (drone ship)	5	16.1%
3	Failure (drone ship)	5	16.1%
4	Success (ground pad)	3	9.7%
5	Controlled (ocean)	3	9.7%
6	Uncontrolled (ocean)	2	6.5%
7	Failure (parachute)	2	6.5%
8	Precluded (drone ship)	1	3.2%

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Key Insights:

- 31 total launches in this 7-year period
- Ground pad landings had 100% success rate (3/3) vs. drone ship at 50% (5/10)
- Early methods (parachute, ocean) were replaced by precision vertical landings
- The 'No attempt' category dominates — reflecting early missions before landing technology matured

SQL Analysis — Key Takeaways

Data-Driven Insights from SQL Analysis:

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- SpaceX has a remarkable 98% mission success rate across 101 launches
- The first-stage landing capability evolved rapidly from 2015 to 2020
- Falcon 9 Block 5 is the workhorse: carries max payload and achieves the most reuse cycles
- NASA (CRS) is the largest customer by payload volume (45,596 kg)
-

Evolution of Landing Capability:

-
- 2010-2014: No landing attempts (parachute recovery or no attempt)
- 2015: First ground pad success (Dec 22) after 2 drone ship failures
- 2016: First drone ship success; success rate reaches ~60%
- 2017-2020: Routine landings; Block 5 boosters flying 7+ times each
-

Ground pad landings have historically been more reliable than drone ship landings (100% vs ~50% in early years)

Section 3: Interactive Map Results — Folium

Folium: All Launch Sites Map

Map 1: SpaceX Launch Site Locations (zoom level 5)

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Launch Site Coordinates:

Launch Site	Latitude	Longitude	State	Orbit Type
CCAFS LC-40	28.5623°N	80.5774°W	Florida	GTO, LEO, ISS
CCAFS SLC-40	28.5632°N	80.5768°W	Florida	LEO, GTO, ISS
KSC LC-39A	28.5733°N	80.6469°W	Florida	Crewed, heavy payload
VAFB SLC-4E	34.6328°N	120.6107°W	California	Polar, SSO

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Geographic Strategy:

- 3 Florida sites clustered within ~8 km — share Cape Canaveral infrastructure
- 1 California site for polar orbits (requires southward trajectory over Pacific)
- All sites are at low latitudes to maximize Earth's rotational velocity advantage
- Florida sites (~28.56°N) are closest to the equator — optimal for equatorial orbits

Folium: Success/Failure Marker Clusters

Map 2: Color-Coded Launch Outcomes

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Marker Color Legend:

-  Green markers = Successful landings (Class = 1)
-  Red markers = Failed landings (Class = 0)
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Observations by Site:

- KSC LC-39A: Predominantly green — highest success rate among all sites
- CCAFS SLC-40: Mixed green/red — reflects long operational history with early failures
- CCAFS LC-40: Mix of outcomes from the early Falcon 9 era
- VAFB SLC-4E: Smaller cluster, mixed outcomes for polar missions
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MarkerCluster Feature:

- Zoomed-out view shows aggregated cluster counts at each site
- Zooming in reveals individual launch markers with success/failure colors
- Interactive popups on each marker provide launch details

Folium: Launch Site Proximity Analysis

Map 3: CCAFS SLC-40 Proximity to Infrastructure

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Feature	Distance	Significance
Coastline	~0.87 km	Rockets launch over open ocean for safety
Highway	~0.61 km	Ground transportation & personnel access
Railway	~1.2 km	Heavy component logistics
Nearest City	~19.1 km	Safety buffer from populated areas

Proximities Analysis

Launch Site Selection Criteria (discovered patterns):

- Coastal: All sites directly on coastline — rockets fly over open water
- Low latitude: Maximizes rotational velocity boost (saves fuel for equatorial orbits)
- Infrastructure access: Close to highways (~0.6 km) and railways (~1.2 km) for logistics
- Safety distance: Minimum ~19 km from populated areas
- East coast for equatorial orbits (Florida), west coast for polar orbits (California)

Section 4: Dashboard Results — Plotly Dash

Dashboard: Launch Success Distribution (All Sites)

Visualization: Pie Chart — Total Successful Launches by Site

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Dashboard Configuration: Dropdown set to 'All Sites'

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Findings:

- The pie chart shows the proportion of successful launches contributed by each site
- KSC LC-39A contributes the largest share of total successes
- CCAFS LC-40/SLC-40 combined represent a significant portion of successes due to high volume
- VAFB SLC-4E has the smallest success contribution (fewer total launches)
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Interactive Features:

- Hover over slices to see exact counts and percentages
- Dropdown selection switches to individual site success/failure view
- Colors differentiate each launch site for easy comparison

Dashboard: Best Launch Site Success Ratio

Visualization: Pie Chart — Success vs. Failure for KSC LC-39A

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Dashboard Configuration: Dropdown set to 'KSC LC-39A'

-

KSC LC-39A Performance:

- Highest success ratio among all launch sites
- Predominantly successful landings (green slice dominates the pie)
- This site handles crewed missions (Crew Dragon) and heavy payloads
- Historic pad — previously used for Apollo and Space Shuttle programs

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Comparison with Other Sites:

- CCAFS LC-40: Lower success ratio due to many early-era launches before landing tech matured
- CCAFS SLC-40: Moderate success ratio with mixed mission types
- VAFB SLC-4E: Mixed results — smaller sample size, specialized polar missions

Dashboard: Payload vs. Launch Outcome

Visualization: Scatter Plot — Payload Mass vs. Class, colored by Booster Version

- Dashboard Configuration: All Sites, Full Payload Range (0-10,000 kg)

Payload Analysis:

Section 3

- Light payloads (<2,000 kg): Mixed outcomes — includes early test flights and small satellite missions
- Medium payloads (2,000-6,000 kg): Generally higher success rates, especially with newer boosters
- Heavy payloads (>6,000 kg): Mostly successful with Block 5 boosters

Build a Dashboard with Plotly Dash

Booster Version Evolution (Color-coded)

- v1.0 (earliest): Exclusively failures — no landing capability
- v1.1: Mostly failures — experimental landing attempts
- FT (Full Thrust): Mixed — transitional period with improving success
- B4 (Block 4): Improved success rate
- B5 (Block 5): Highest success rate — current operational booster with proven reusability
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Key Insight: Booster version is the strongest predictor of landing success, followed by payload mass

Section 5: Predictive Analysis Results

Classification Model Accuracy Comparison

Model Performance Summary:

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Model	CV Accuracy (10-fold)	Test Accuracy	Best Hyperparameters
Logistic Regression	84.64%	83.33%	C=0.01, penalty=l2, solver=lbfgs
SVM (SVC)	84.82%	83.33%	C=1.0, gamma=0.0316, kernel=sigmoid
Decision Tree	88.93%	83.33%	criterion=gini, max_depth=2, min_leaf=2
KNN	84.82%	83.33%	n_neighbors=10, p=1, algorithm=auto

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Analysis:

- All 4 models achieved identical test accuracy: 83.33% (15/18 correct predictions)
- Decision Tree achieved the highest cross-validation accuracy: 88.93%
- The small test set (18 samples) limits discriminatory power between models
- 10-fold CV on training data provides more reliable performance estimates
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Decision Tree has the best overall performance based on cross-validation accuracy

Confusion Matrix — All Models (Identical)

All 4 models produced the same confusion matrix on the 18-sample test set:

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	Predicted: Did Not Land (0)	Predicted: Landed (1)
Actual: Did Not Land (0)	3 (True Negative)	3 (False Positive)
Actual: Landed (1)	0 (False Negative)	12 (True Positive)

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Performance Metrics:

- Accuracy: 83.33% (15/18 correct)
- Precision (for landing prediction): 80% (12/15) — some false positives
- Recall (for landing prediction): 100% (12/12) — never misses a successful landing
- Specificity: 50% (3/6) — struggles to identify failures
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Error Analysis:

- 3 False Positives: Model predicted successful landing when booster actually failed to land
- 0 False Negatives: Model never failed to predict an actual successful landing
- The models tend to be 'optimistic' — biased toward predicting success
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For example, the model correctly predicted that the booster would land successfully 12 out of 12 times, but it also incorrectly predicted that the booster would land successfully 3 out of 6 times.

Predictive Analysis — Additional Insights

Why All Models Converged to the Same Test Accuracy:

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- Small dataset: Only 90 samples total (72 train, 18 test)
- Class imbalance: 66.67% success rate means a naive 'always predict success' baseline = 66.67%
- All models learned similar decision boundaries given the limited data

Section 6

- The 3 misclassifications are consistently the same borderline cases across all models

Predictive Analysis (Classification)

Feature Importance Observations:

-
- Flight number (proxy for time/experience) is a strong predictor
- Booster version/block number captures technology evolution
- Payload mass has a moderate negative effect on landing success
- Launch site and orbit type contribute to prediction via one-hot encoding
-

Recommendations for Improvement:

-
- Collect more data as new launches occur (dataset only goes to Nov 2020)
- Consider ensemble methods (Random Forest, Gradient Boosting) for better performance
- Explore feature selection to reduce dimensionality (83 features for 90 samples is high)

Model Selection & Comparison

Cross-Validation Accuracy Comparison:

•

Model	CV Score	Test Score	Pros	Cons
Decision Tree	88.93%	83.33%	Highest CV, interpretable	Prone to overfitting
SVM (Sigmoid)	84.82%	83.33%	Robust to outliers	Kernel choice sensitive
KNN (k=10)	84.82%	83.33%	Simple, non-parametric	Sensitive to feature scaling
Logistic Reg.	84.64%	83.33%	Fast, probabilistic output	Linear decision boundary

•

Recommendation:

- Decision Tree is the top performer by cross-validation (88.93%)
- For production use, Logistic Regression offers the best balance of simplicity and performance
- All models are viable given the small performance differences
- Ensemble methods would likely outperform all individual models on larger datasets

Business Impact & Cost Analysis

Launch Cost Implications:

-
- SpaceX Falcon 9 launch cost: ~\$62 million (with first-stage recovery)
- Without recovery: ~\$165 million+ (comparable to competitors like ULA, Arianespace)
- Cost savings from reuse: ~\$103 million per launch (\$165M - \$62M)
-

Prediction Value for Competitors:

-
- If a model predicts landing failure → SpaceX launch effectively costs ~\$165M → Competitors can bid competitively
- If a model predicts landing success → SpaceX launch costs ~\$62M → Hard to undercut
-

Risk Assessment:

-
- Model's false positive rate (predicting success when failure occurs) could lead to underestimating SpaceX's cost
- The model never produces false negatives — it correctly identifies all successful landings
- For conservative bidding strategy, apply a confidence margin to model predictions
-

With 83.33% accuracy, competitors can make informed pricing decisions for ~5 out of 6 launch bids

Conclusions

1. Landing Success Has Dramatically Improved

- From 0% in 2010-2013 to 95%+ by 2020, driven by iterative hardware and software improvements
-

2. Key Success Factors Identified

- Booster version (Block 5 = highest success), flight experience, payload mass, orbit type, and launch site
-

3. KSC LC-39A is the Most Successful Launch Site

- Highest success ratio, handles crewed missions and heavy payloads
-

4. All ML Models Achieve 83.33% Test Accuracy

- Decision Tree leads in cross-validation (88.93%); all models have identical confusion matrices
-

5. Model Bias: Optimistic Predictions

- 100% recall but only 50% specificity — models are better at predicting success than failure
-

6. Reusability is the Key Cost Differentiator

- Accurate landing prediction directly translates to \$103M per-launch cost estimation
-

7. Geographic Factors Matter

Appendix — Project Resources

GitHub Repository:

<https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor>

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Notebooks:

- 1. Data Collection (API): jupyter-labs-spacex-data-collection-api-v2.ipynb
- 2. Data Collection (Web Scraping): jupyter-labs-webscraping.ipynb
- 3. Data Wrangling: labs-jupyter-spacex-Data wrangling-v2.ipynb
- 4. EDA with Visualization: jupyter-labs-eda-dataviz-v2.ipynb
- 5. EDA with SQL: jupyter-labs-eda-sql-coursera_sqlite.ipynb
- 6. Interactive Maps (Folium): lab-jupyter-launch-site-location-v2.ipynb
- 7. Machine Learning Prediction: SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb
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Dashboard:

- spacex-dash-app.py — Plotly Dash interactive dashboard
-

Datasets:

- dataset_part_1.csv, dataset_part_2.csv, dataset_part_3.csv, spacex_launch_dash.csv, spacex_web_scraped.csv
-

Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Folium, Plotly Dash, SQLite, BeautifulSoup, Requests

Thank You

Questions & Discussion

Manoj De Silva

<https://github.com/mdes5793/SpaceX-Launch-Cost-Predictor>